Breast Ultrasound Image Classification Using Deep Learning and Imbalance Handling Techniques

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1 Project Overview

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide. Early detection plays a critical role in improving treatment outcomes and survival rates. Medical imaging techniques such as ultrasound are widely used for breast cancer diagnosis because they are non-invasive, accessible, and cost-effective.

Recent advancements in deep learning have significantly improved medical image analysis. Convolutional Neural Networks (CNNs) have demonstrated strong performance in tasks such as tumor detection and classification.

This project focuses on developing a deep learning-based system to classify breast ultrasound images into three categories: benign, malignant, and normal. The MobileNetV2 architecture is used through transfer learning to build an efficient classifier.

Additionally, this project investigates different techniques to handle class imbalance present in the dataset. Several experiments were conducted including a baseline model, class-weight balancing, data augmentation, and SMOTE-based oversampling.

2 Objective

The main objective of this project is to develop an automated deep learning model for breast ultrasound classification.

The specific goals include:

- Implementing a CNN classifier using MobileNetV2 transfer learning

- Training a baseline model using the imbalanced dataset
- Applying class-weight balancing to address class imbalance
- Applying data augmentation to increase dataset diversity
- Applying SMOTE oversampling for feature-level balancing
- Comparing the performance of different imbalance handling techniques

3 Dataset

This project uses the publicly available Breast Ultrasound Images Dataset (BUSI). The dataset contains ultrasound images collected from female patients aged between 25 and 75 years.

The images are categorized into three classes:

- Benign
- Malignant
- Normal

Each image is accompanied by a segmentation mask highlighting the tumor region. For this classification task, only the ultrasound images were used and the mask images were removed during preprocessing.

3.1 Dataset Statistics

The dataset contains a total of 780 ultrasound images distributed across three classes.

Class	Number of Images
Benign	437
Malignant	210
Normal	133

Table 1: BUSI Dataset Distribution

The dataset is imbalanced, with significantly more benign samples compared to malignant and normal samples.

The dataset was divided into two subsets:

- Training Set: 80%
- Test Set: 20%

4 Methodology

To study the effect of class imbalance on model performance, several techniques were explored. The overall workflow of the proposed system includes dataset preprocessing, feature extraction using MobileNetV2, and applying different imbalance handling techniques.

4.1 Baseline Model

The baseline model was trained on the original dataset without applying any class imbalance handling techniques. This experiment provides a reference point for evaluating the effectiveness of other balancing methods.

4.2 Class Weight Balancing

Class weights were calculated based on the distribution of the dataset and applied during model training. This approach increases the penalty for misclassifying minority classes such as malignant and normal samples.

4.3 Data Augmentation

Data augmentation techniques were applied to increase the diversity of training data. The augmentation methods included small rotations, zooming, and horizontal flipping. These transformations help improve model generalization and reduce overfitting.

4.4 SMOTE Oversampling

The Synthetic Minority Oversampling Technique (SMOTE) was applied to feature vectors extracted from MobileNetV2. SMOTE generates synthetic samples for minority classes, resulting in a balanced feature space that improves classification performance.

5 Data Preparation

Several preprocessing steps were applied before training the model:

- All ultrasound images were resized to 128×128 pixels
- Pixel values were normalized to the range [0,1]
- Segmentation mask images were removed from the dataset
- The dataset was divided into training and testing subsets

These preprocessing steps improve model convergence and reduce computational complexity.

6 Model Architecture

The classification model uses MobileNetV2, a lightweight convolutional neural network designed for efficient deep learning applications.

Transfer learning was applied by initializing MobileNetV2 with pretrained ImageNet weights. The convolutional backbone was frozen during training to preserve previously learned feature representations.

The model architecture includes:

- Pretrained MobileNetV2 feature extractor
- Global Average Pooling layer
- Fully connected Dense layer
- Softmax output layer with three classes

The output layer predicts the probability of each image belonging to the benign, malignant, or normal class.

7 Training Configuration

The model was trained using the following configuration.

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001 (default Adam learning rate)
Loss Function	Categorical Crossentropy
Batch Size	32
Epochs	20
Input Size	128×128

Table 2: Training Hyperparameters

8 Evaluation Metrics

Model performance was evaluated using the following metrics:

- Accuracy

- Precision
- Recall
- F1 Score

These metrics provide a comprehensive evaluation of the classification performance.

9 Results

The performance comparison of the different techniques is shown below.

Method	Accuracy
Baseline Model	76.3%
Class Weight Balancing	80.7%
Data Augmentation	80.1%
SMOTE Oversampling	97.4%

Table 3: Performance Comparison

The results show that applying imbalance handling techniques improves classification performance. Among the tested methods, SMOTE-based feature balancing achieved the highest accuracy.

9.1 Discussion of Results

The experimental results demonstrate the impact of different class imbalance handling techniques on model performance.

The baseline model achieved an accuracy of approximately 76.3%. Although the MobileNetV2 architecture provides strong feature extraction through transfer learning, the imbalanced dataset causes the model to learn dominant patterns from the majority class (benign images).

Applying class-weight balancing improved the model performance to 80.7%. This improvement occurs because higher penalties are assigned to errors made on minority classes during training, encouraging the model to learn more representative features for malignant and normal samples.

Data augmentation produced a similar accuracy of 80.1%. Augmentation increases dataset diversity by applying transformations such as rotation, zooming, and flipping, which helps improve generalization and reduces overfitting.

The highest accuracy (97.4%) was achieved using the SMOTE-based approach. In this implementation, feature vectors extracted from the pretrained MobileNetV2 network were used as input for SMOTE. Since deep convolutional networks generate highly discriminative feature representations, applying SMOTE in the feature space allows synthetic samples to be generated between similar data points. This results in a more balanced and separable feature distribution across the three classes.

Consequently, the classifier trained on the SMOTE-balanced feature space is able to learn clearer decision boundaries between benign, malignant, and normal images, leading to significantly improved classification accuracy.

10 Conclusion

This project demonstrates the effectiveness of deep learning for breast ultrasound image classification. Transfer learning with MobileNetV2 enables strong feature extraction even with limited medical datasets.

Different imbalance handling techniques were explored, including class-weight balancing, data augmentation, and SMOTE oversampling. The results show that these techniques significantly improve model performance compared to the baseline model.

Future work could explore advanced architectures such as EfficientNet, attention-based models, or explainable AI techniques to further improve diagnostic accuracy.